

Identification of solar panels in Stuttgart Using Machine learning

Remote Sensing Studio 2024

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Table of contents

01

Introduction

What is the project about?

02

Project Design

Which architecture was used?

05

Future Outlook

What could be better?

03

Results

Comparison of different tries.

04

Issues

Problems faced.

06

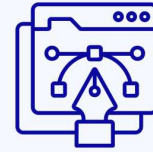
Conclusions

01 Introduction

This project is focused in designed and training a deep learning architecture to identificate **Solar Panels in Stuttgart City** from aerial imagery.

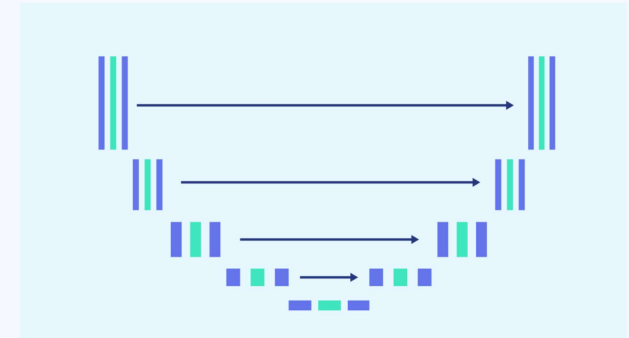


02 Project Desing U-Net architecture

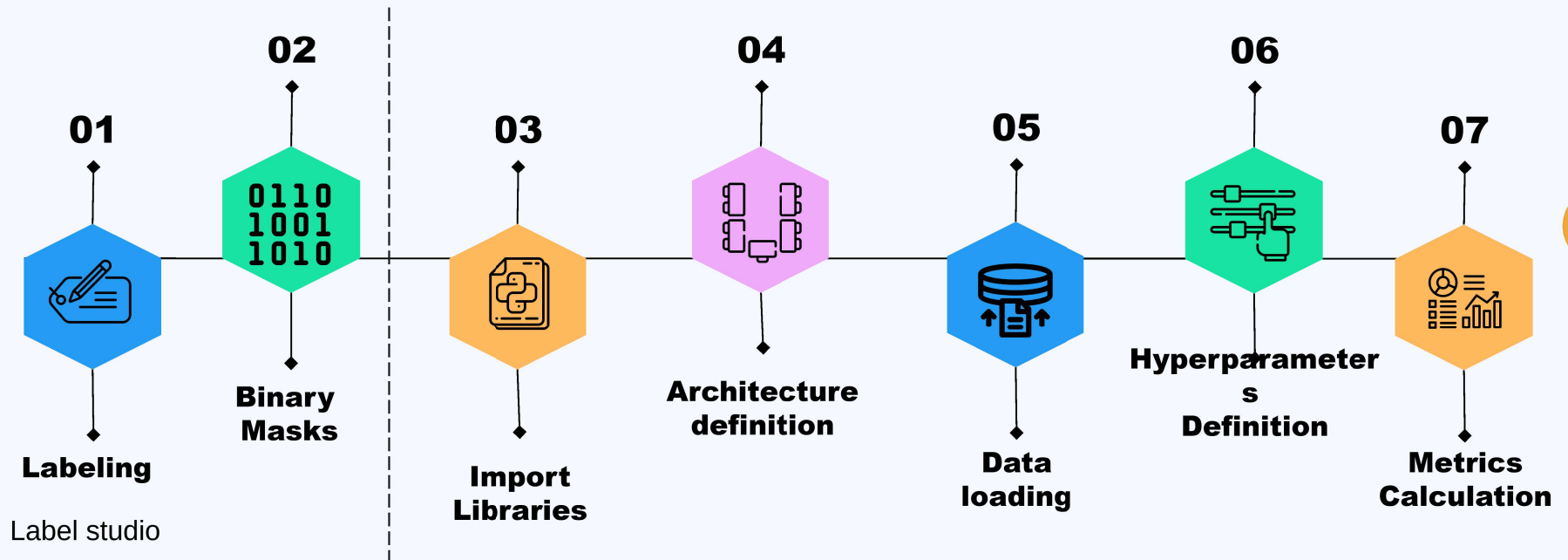


U-Net's design is particularly well-suited for image segmentation due to its ability to capture both context and detailed localization information through its encoder-decoder architecture and skip connections.

This flexibility and effectiveness make U-Net a valuable tool in fields that require precise image analysis, remote sensing and also medical approaches.



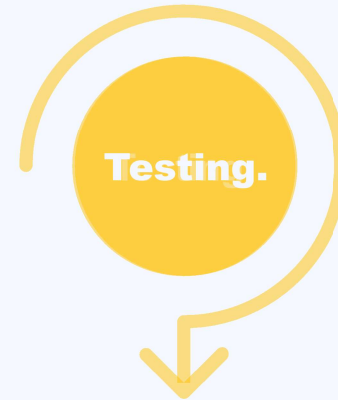
Pipeline Validation



Validation and Testing



- Parameters selection and Tuning
- Lockover throught the model for revition
- Confussion Matrix



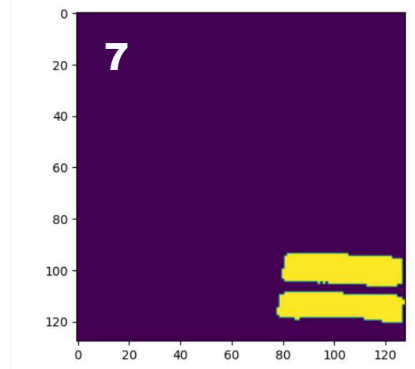
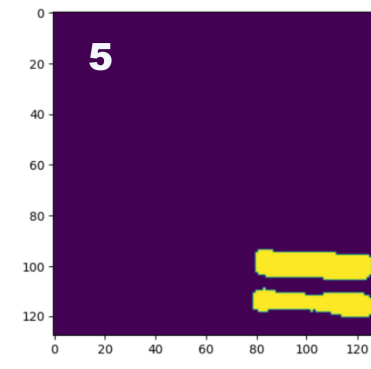
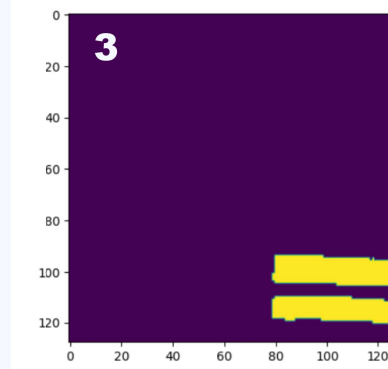
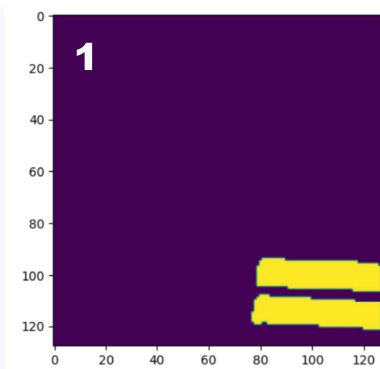
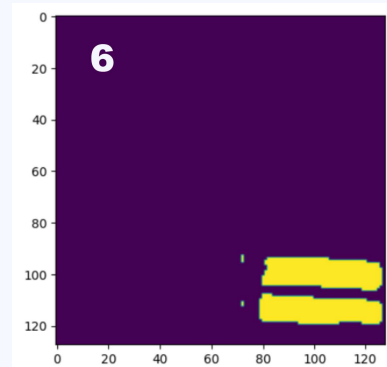
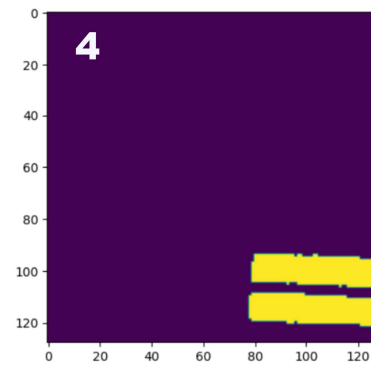
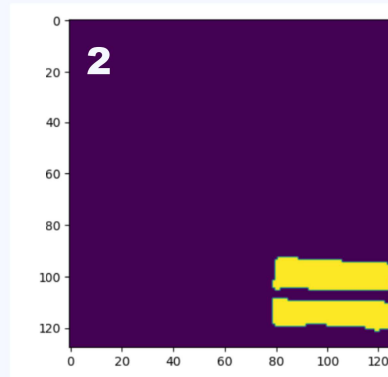
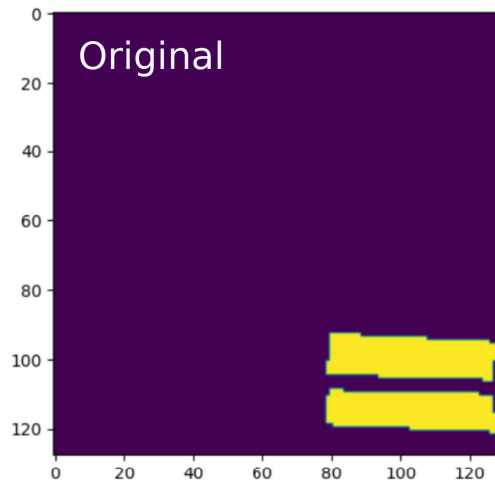
- Run model on unseen data
- Final Result

03

Results

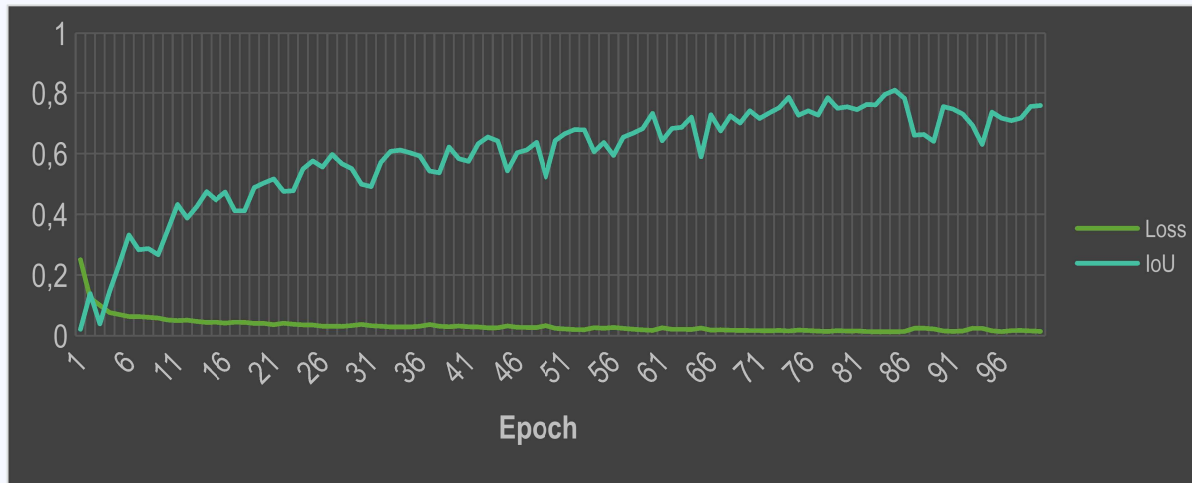


Predictions on training stage



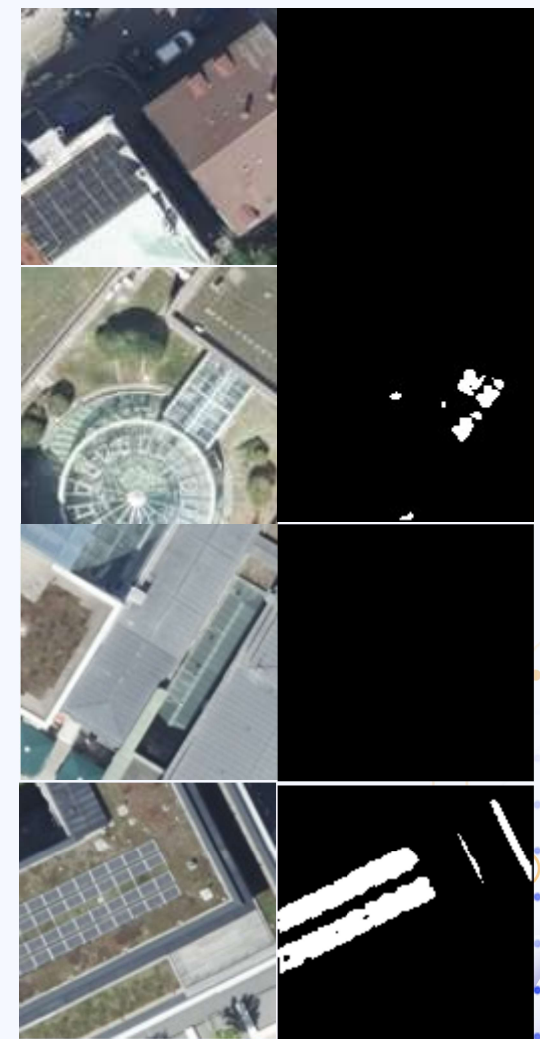
Exp. 1

Total Images	Mask	No Mask	Lr	Batch	Epoch
304	41	59	0,001	8	100



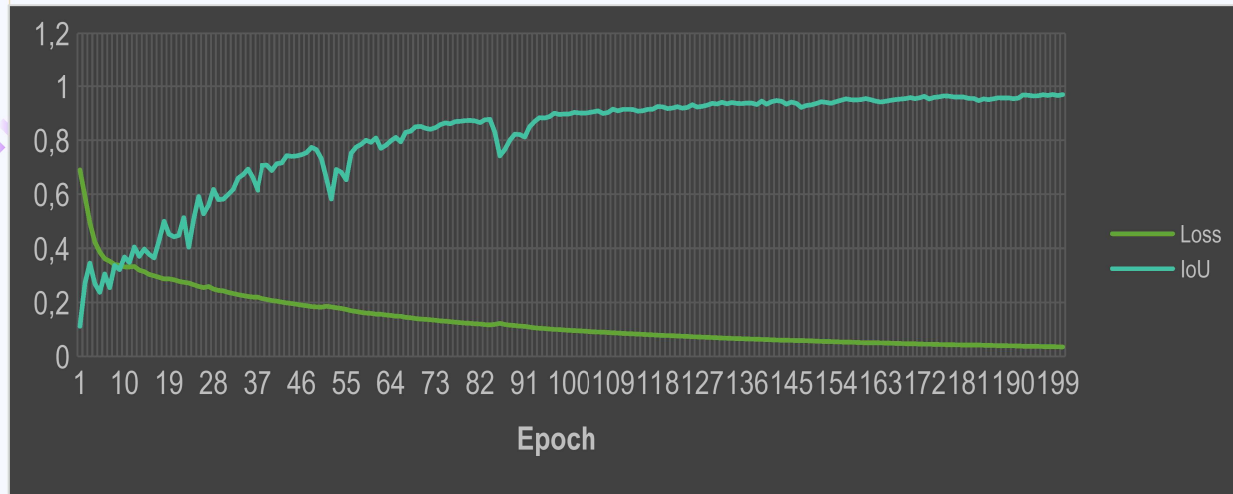
Kappa Coef.	Avg Acc	Ave Prec	Avg Recall	Avg F1Score
0,5933	0,9792	0,7391	0,6849	0,7045

	Ground- Truth	
Prediction	14066	4991
	4645	43504



Exp . 2

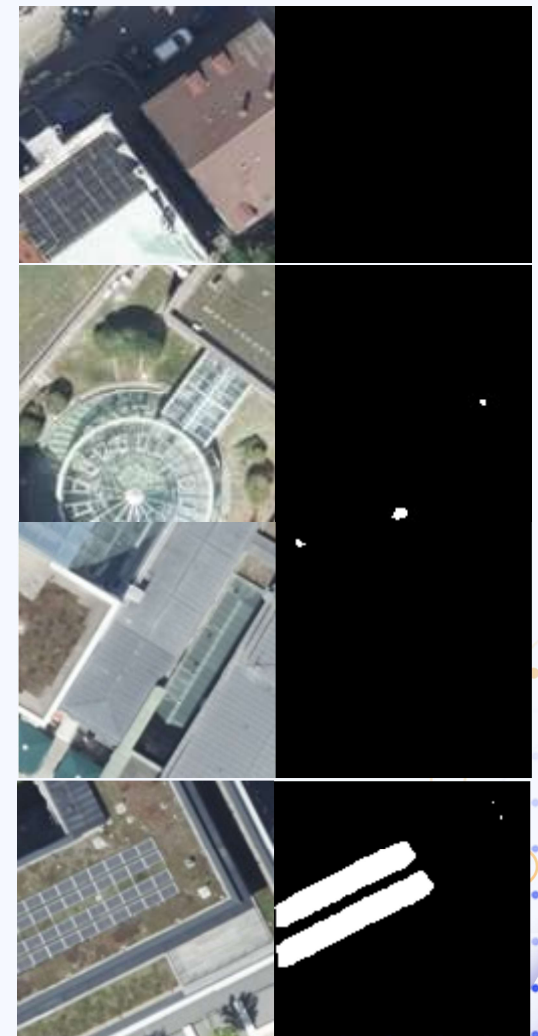
Total Images	Mask	No Mask	Lr	Batch	Epoch
304	41	59	0,0001	40	200



Kappa Coef.	Avg Acc	Ave Prec	Avg Recall	Avg F1Score
0,6532	0,9789	0,8006	0,5805	0,6637

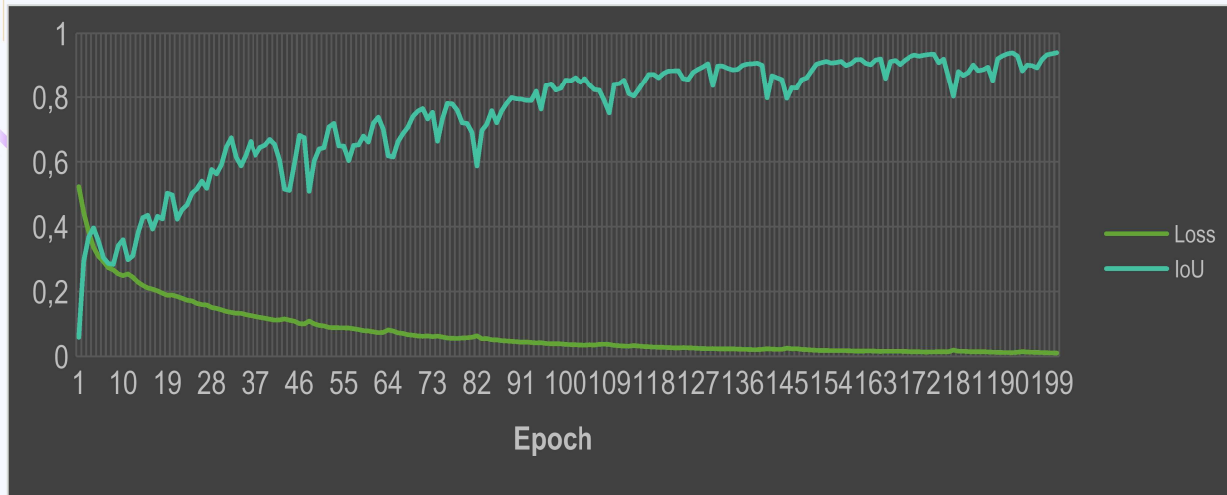
	Ground- Truth	
Prediction	12118	6939
	2764	436931

10



Exp . 3

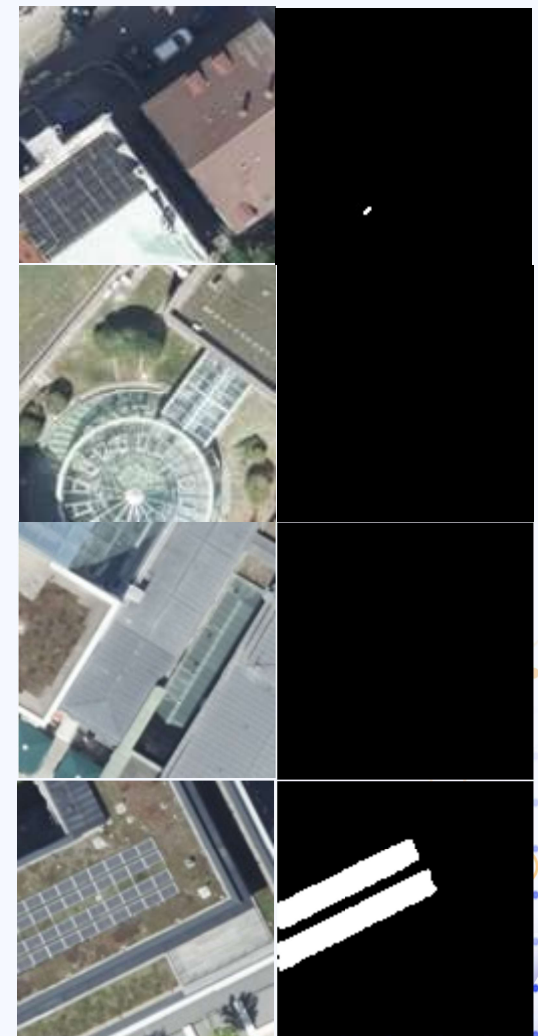
Total Images	Mask	No Mask	Lr	Batch	Epoch
527	32	68	0,0001	40	200



Kappa Coef.	Avg Acc	Ave Prec	Avg Recall	Avg F1Score
0,7489	0,9836	0,8142	0,733	0,7571

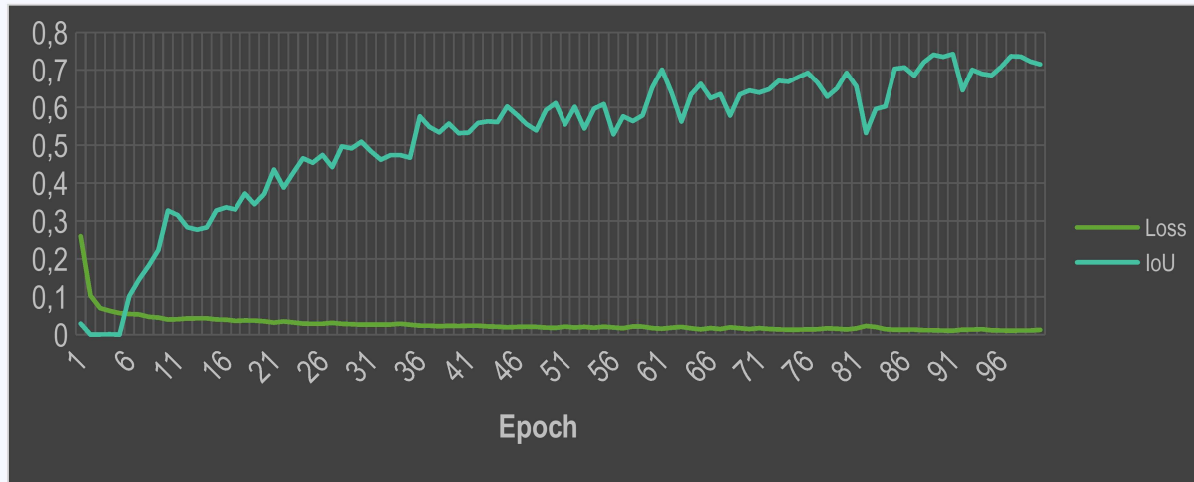
	Ground-Truth	
	Prediction	
	22765	6663
	5047	702805

11



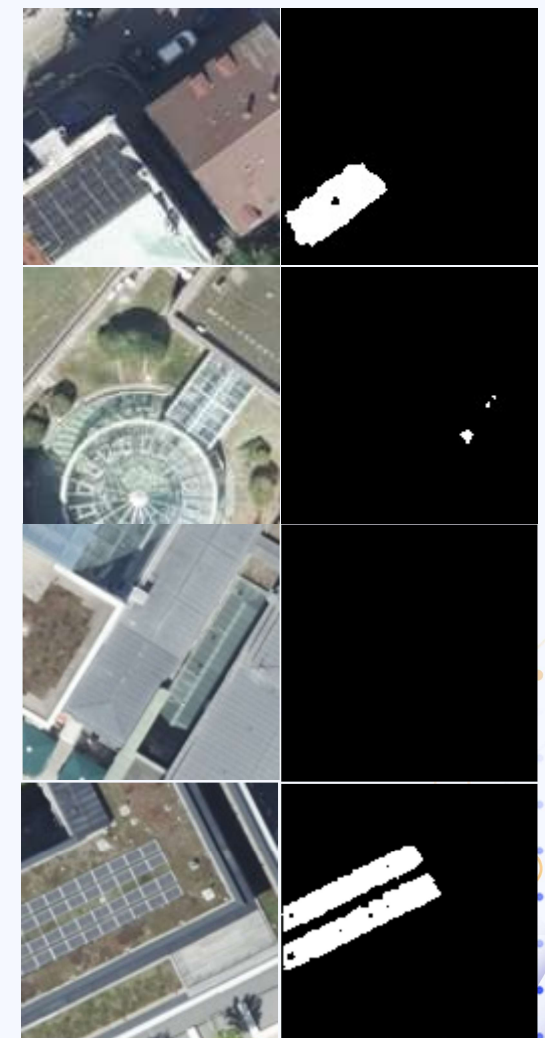
Exp. 4

Total Images	Mask	No Mask	Lr	Batch	Epoch
527	32	68	0,001	8	100



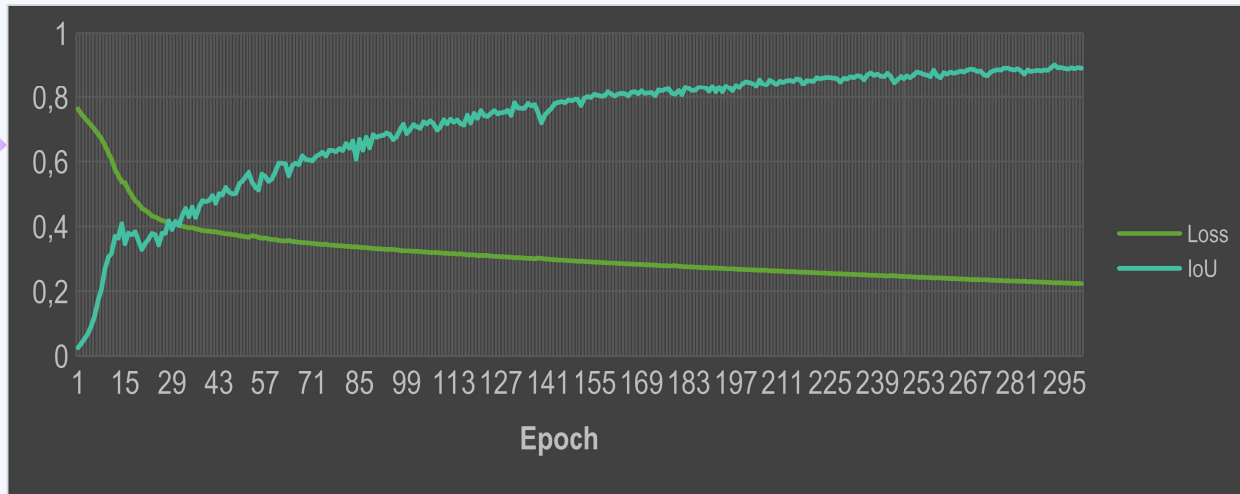
Kappa Coef.	Avg Acc	Ave Prec	Avg Recall	Avg F1Score
0,5879	0,9779	0,8308	0,5007	0,5972

	Ground- Truth	
Prediction	16035	13393
	2549	705303



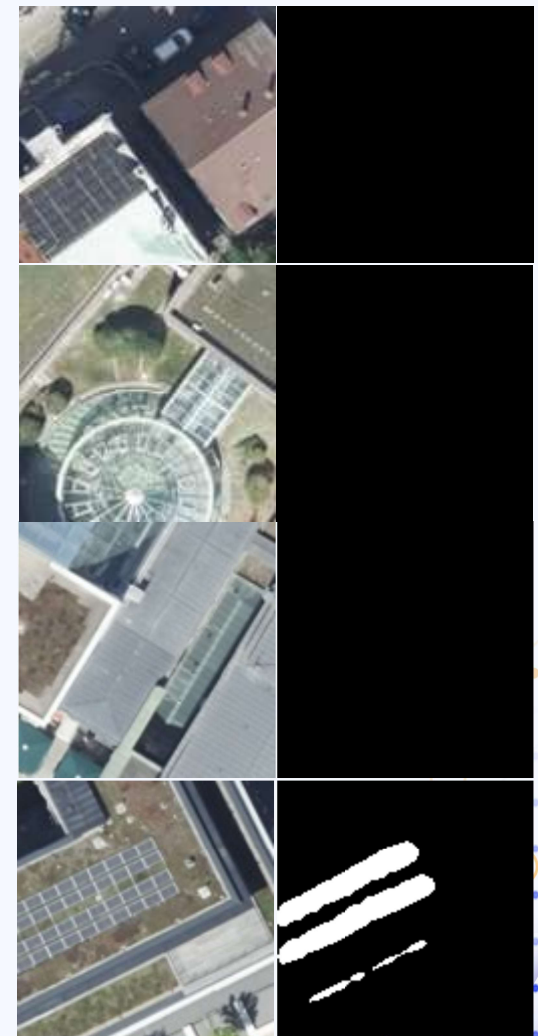
Exp. 5

Total Images	Mask	No Mask	Lr	Batch	Epoch
527	32	68	0,00001	60	300



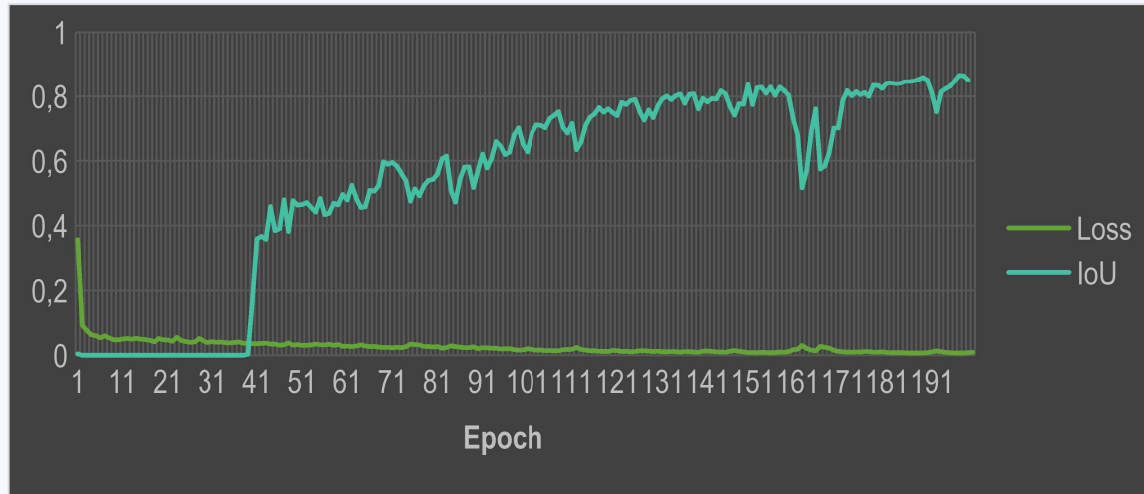
Kappa Coef.	Avg Acc	Ave Prec	Avg Recall	Avg F1Score
0,5357	0,9738	0,7515	0,4569	0,5474

	Ground- Truth	
Prediction	14520	14908
	4178	703674



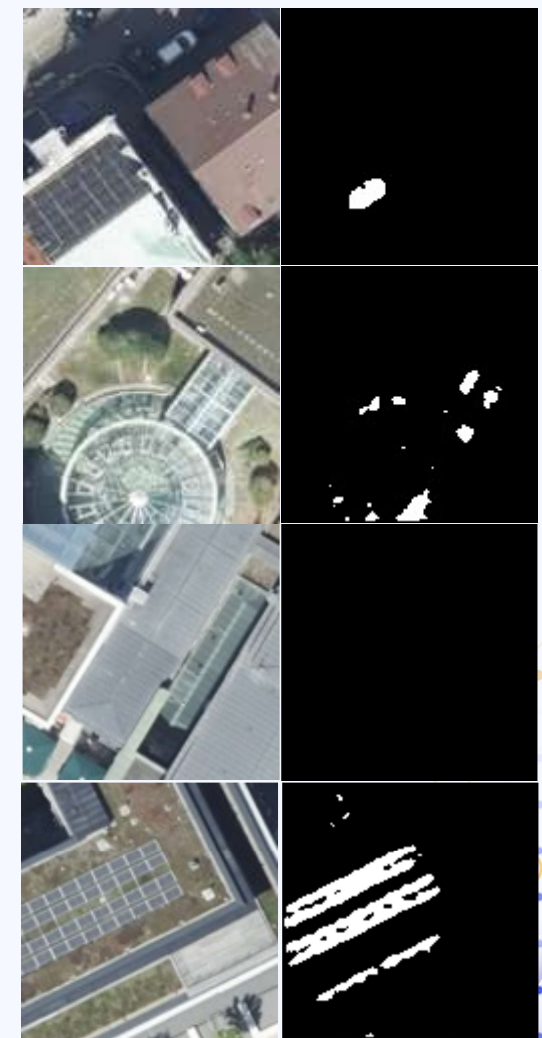
Exp . 6

Total Images	Mask	No Mask	Lr	Batch	Epoch
527	32	68	0,01	40	200



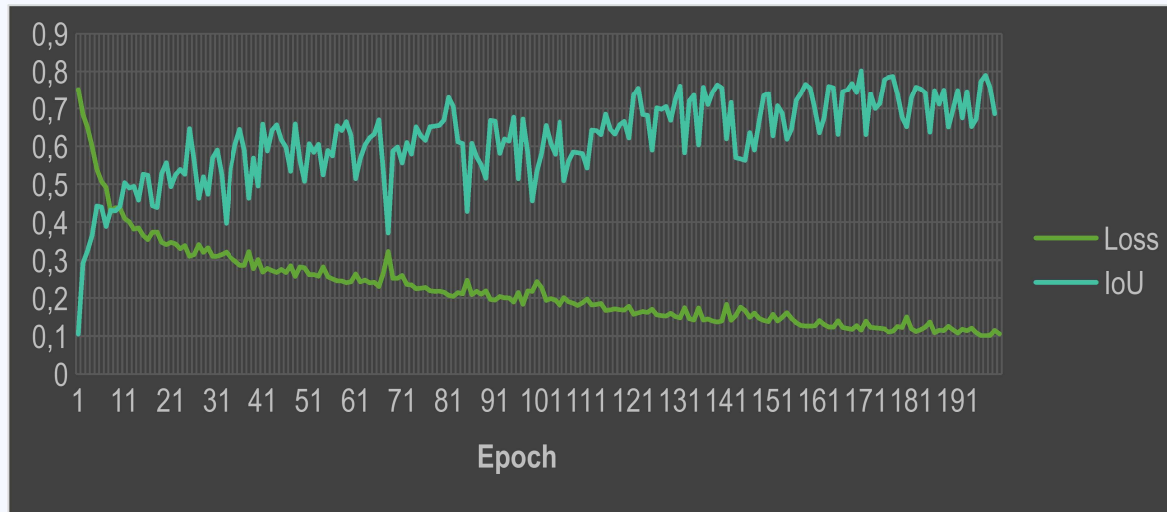
Kappa Coef.	Avg Acc	Ave Prec	Avg Recall	Avg F1Score
0,704	0,9799	0,753	0,6924	0,7142

	Ground-Truth	
	Prediction	
	21326	8102
	6450	701402



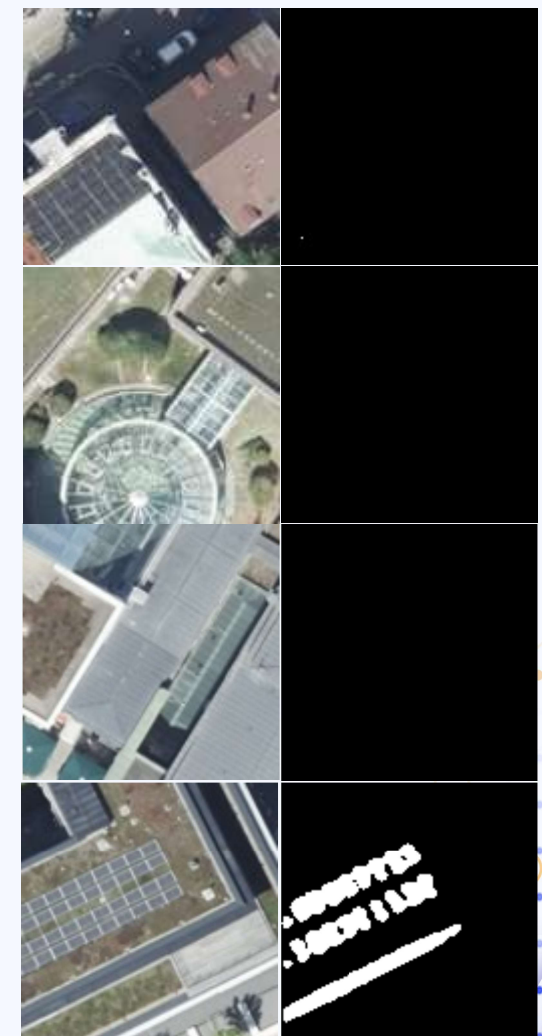
Exp. 7

Total Images	Mask	No Mask	Lr	Batch	Epoch
161	100	0	0,0001	40	200



Kappa Coef.	Avg Acc	Ave Prec	Avg Recall	Avg F1Score
0,6709	0,9799	0,8406	0,5902	0,6807

	Ground- Truth	
Prediction	18358	11070
	3529	704323

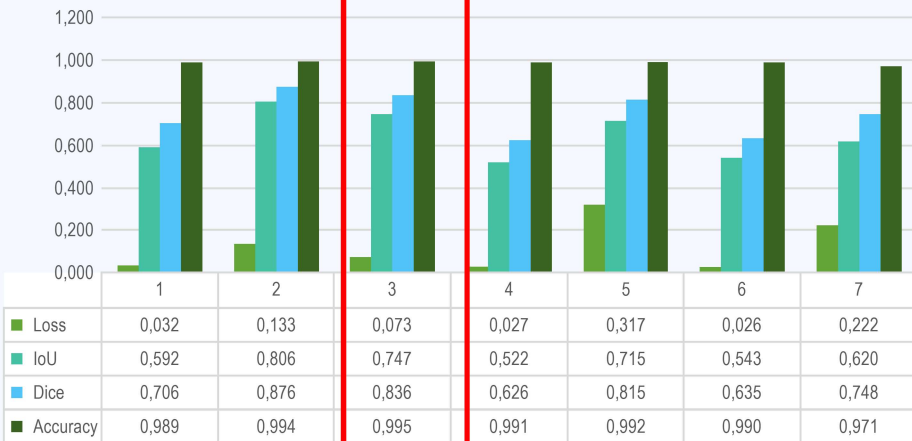


Comparison

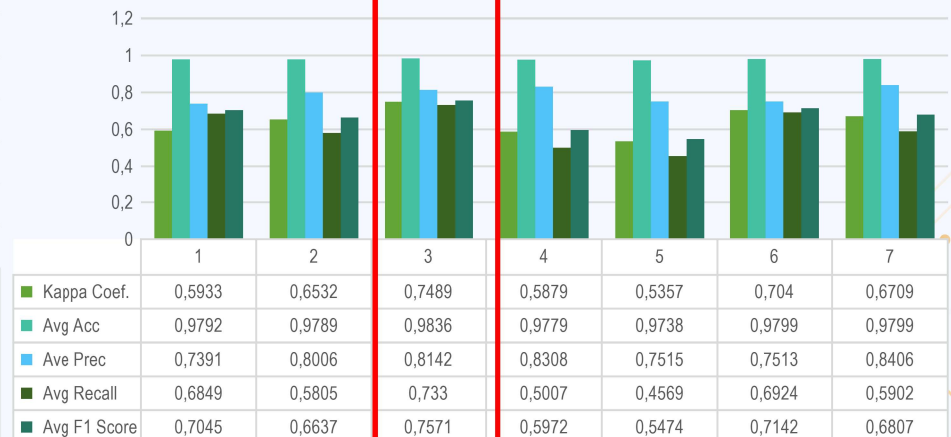


	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5	Exp. 6	Exp. 7
Lr	0,001	0,0001	0,0001	0,001	0,00001	0,01	0,0001
batch	8	40	40	8	60	40	40
Epoch	100	200	200	100	300	200	200

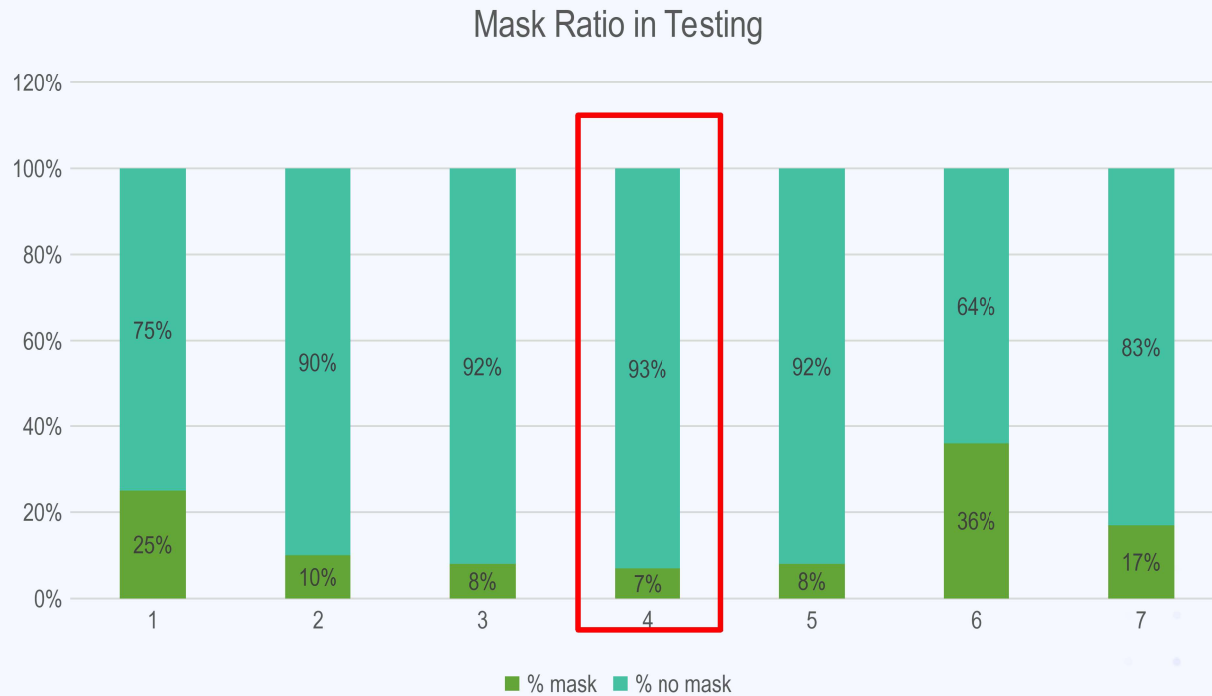
Training Metrics



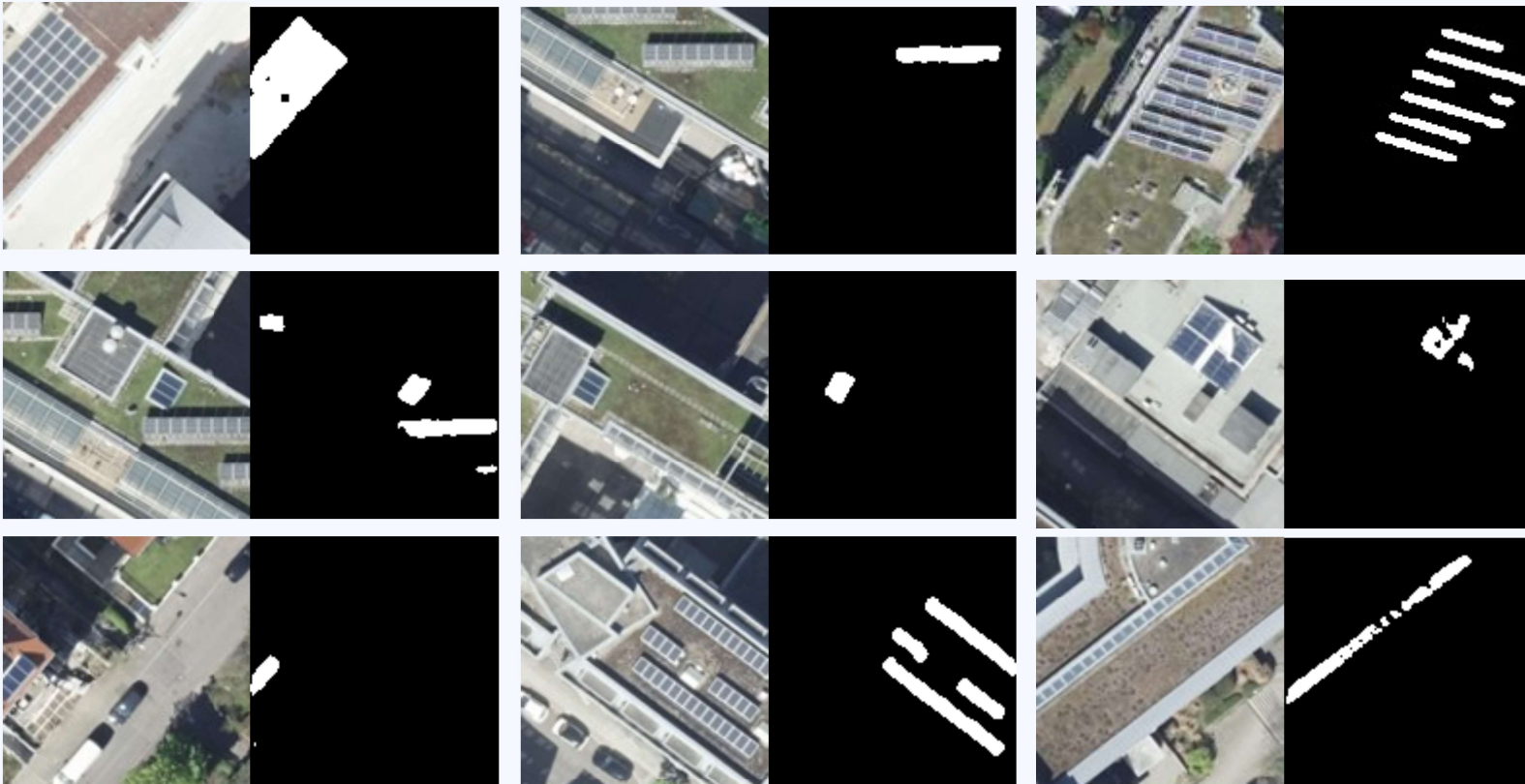
Validation Result



Comparison



Exp. 3 Testing result



04. Issues



Masks Match

Masks Loading and reading

Number of convolutions

GPU issues

Few images with solar panels



05 Future Outlook

More Computation power / timeF

Try with more and
different parameters

Try different
type of labeling

Try with different
dataset size

Try with another
architecture



06 Conclusions

- Huge range of applications using deep learning.
- The implementation is limited to the computation power.
- To work with appropriate metrics is important to evaluate the performance of our experiments and chose the best result.

